

The Transfer Function in TF/RL

From CT intensity to rendered anatomy, goals, and policy actions

Kevin Kunkel · Master's thesis · Technical note

Note: This note describes the transfer-function representation currently used by the viewer and reinforcement-learning pipeline. It is a constrained semantic parameterization, not an arbitrary point-by-point curve editor.

1. Purpose

A direct volume renderer maps a three-dimensional scalar field to an image. In the CT setting, each voxel carries an intensity measured in Hounsfield Units (HU). A transfer function (TF) maps that intensity to colour and opacity:

$$T : HU \rightarrow (r, g, b, \alpha)$$

Opacity controls how strongly a sample contributes to the image. Colour controls its appearance. The same opacity curve can produce very different images on two patients: tissue in front of a target can hide it. Therefore, this project does not optimise opacity in isolation. It measures what the completed rendering actually shows.

2. Current Representation

The current TF uses four Gaussian peaks over the CT intensity range. Each peak represents one semantic material group:

Class	Centre (HU)	Role
lungs	-800	lung parenchyma
soft	40	organs and muscle together
vessels	300	contrast-enhanced vessels
skeleton	900	bone

For peak i , opacity contribution around intensity x is modelled as:

$$\alpha_{i(x)} = h_i \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right)$$

where μ_i is peak centre, σ_i is width, and h_i is height. Colour comes from the peak RGB values. The renderer combines contributions from all peaks into lookup tables before compositing samples through the volume.

The full internal parameter vector contains six values per peak:

centre μ_i · **width** σ_i · **height** h_i · **red, green, blue** (r_i, g_i, b_i)

Thus, four peaks give 24 internal values. Peak centres remain fixed at the anatomical positions above. The policy controls the other 12 parameter groups:

Controllable group	Meaning
width	how wide an intensity neighbourhood contributes

Controllable group	Meaning
height	opacity strength of a peak
RGB colour	colour intensity; one action controls the group mean

Parameters are represented internally in a normalized range. `transfer.py` converts them to renderable transfer-function values and keeps peak semantics stable across commands and policy outputs.

Design choice: Fixed centres reduce action-space size and preserve semantic meaning. The tradeoff is expressiveness: current system cannot move peak centres or draw arbitrary piecewise-linear control points.

3. Colour Preservation

RGB is represented as one controllable group per peak, but the implementation does not force all three channels to become identical. `apply_controllable` sets the group mean while preserving each channel's offset from that mean:

$$r'_i = v_i + (r_i - |(r_i, g_i, b_i)|)$$

with equivalent equations for green and blue. Values are clipped to the valid normalized range after reconstruction.

This preserves peak hue when a policy changes colour intensity. Without this offset-preserving decode, policy outputs collapse to grey and can become visually identifiable in a supposedly blind preference comparison.

4. From Transfer Function to Image

Rendering samples the volume along viewing rays. For each sample, the TF gives opacity and colour. Samples are composited front-to-back. A simplified update is:

$$C_{\text{out}} = C_{\text{in}} + (1 - A_{\text{in}})\alpha C$$

\

$$A_{\text{out}} = A_{\text{in}} + (1 - A_{\text{in}})\alpha$$

where C is sample colour, α sample opacity, and A accumulated opacity. The order matters. A high-opacity skin or muscle layer can reduce later contributions from bone, even when the bone peak itself is high.

Caveat: Changing one peak has side effects. Increasing bone opacity can alter occlusion, brightness, and coverage of other classes. This is why a TF-curve area metric alone is insufficient.

5. Visibility Measurement

`visibility.py` provides a fast approximation of what the rendered image shows. For each volume, it resamples views into a coarse front-to-back grid. Each sample has an anatomical class label where TotalSegmentator masks are available. The same transfer-function lookup and compositing logic then estimates class contribution.

Raw measured classes are:

skeleton · lungs · organs · muscle · vessels · coverage

The user-facing goal classes are **skeleton**, **lungs**, **soft**, and **vessels**. **soft** combines organs and muscle because their intensity ranges overlap and a one-dimensional TF cannot reliably separate them.

For every class, measurement produces visibility and brightness. Coverage tracks whether the rendered frame contains meaningful accumulated opacity. The approximation is fast enough for reinforcement-learning loops and was validated against label-masked VTK renders. Colour-blackening preserves opacity and occlusion, isolating a class’s image contribution more faithfully than deleting its voxels.

6. Natural Language to TF Goal

An instruction is parsed into a command, then compiled into a goal vector. For example:

```
show more bone → increase skeleton visibility
brighten the skeleton → increase skeleton brightness
more bone, less soft tissue → compound visibility goal
```

The goal stores, per user-facing class:

- target change in logarithmic visibility;
- mask indicating which classes are mentioned;
- target brightness change and mask;
- tolerance for unmentioned-class drift.

Logarithmic visibility makes verbal changes approximately multiplicative. A positive target asks for more visible anatomy; a negative target asks for less. The keep term prevents unnecessary changes to unmentioned classes but allows small side effects caused by occlusion.

7. Attainment and RL Interface

Let D_{start} be distance from the initial rendering to the instruction goal, and D_{final} distance after applying a candidate TF. Reported attainment is:

$$A = 1 - \frac{D_{\text{final}}}{D_{\text{start}}}$$

Interpretation:

Attainment	Meaning
1	goal reached
0	no better than leaving TF unchanged
< 0	candidate made goal worse

The one-shot policy observes 57 values:

goal vector (16) + histogram (16) + current visibility (4) + current brightness (4) + reachable ceilings (4) + current controllable values (12) + coverage (1)

It outputs 12 values, one per controllable width, height, or RGB group. In absolute mode, outputs are new group values. Experimental residual mode treats outputs as bounded adjustments from the current TF. Both modes remain one-step: the policy proposes one complete TF, without iterative policy actions.

The same attainment objective evaluates:

- exact command application;
- coordinate hill-climb search;
- the learned policy;
- policy plus measured refinement.

This shared objective prevents the application comparison panel from silently using a different ruler than the thesis evaluation.

8. What Current TF Does Not Yet Do

Caveat: Current representation does not support arbitrary control-point insertion, direct line-segment editing, or moving Gaussian centres. It also does not learn from human preferences yet. The preference collector is implemented, but the reported policy is trained against the validated analytical objective.

The design prioritizes semantic, scan-aware instruction following over maximum TF expressiveness. Future work can add a richer piecewise-linear parameterization, DINOv2-based visual reranking, or a human-preference reward model after the measurement and one-shot policy interfaces are established.

9. Representation Tradeoff and Future Hybrid Design

The original task description assumes a conventional piecewise-linear VTK transfer function: consecutive control points define opacity and colour ramps over the intensity histogram. That representation and the current Gaussian representation serve different purposes.

Property	Piecewise-linear TF	Semantic Gaussian TF
Expressiveness	arbitrary ramps, points, and sharp transitions	four fixed semantic peaks
User control	precise expert curve editing	compact tissue-level commands
RL interface	variable or large action space	fixed 12-value action
Language mapping	ambiguous point selection	bone/lung/vessel/soft targets
Generalisation	can represent unusual materials	depends on fixed peak centres
Evaluation	point or curve distance is insufficient	scan-aware visibility attainment

Gaussian parameterization is therefore a research instrument, not a claim that piecewise-linear transfer functions are obsolete. It reduces search space, preserves semantic tissue names, and makes fixed-dimensional policy possible. Cost: reduced curve expressiveness and dependence on anatomical peak assumptions.

Design choice: Most current contribution lies at intersection of two designs: semantic transfer-function parameterization and RL policy learning. Policy question is tractable because TF representation is constrained; TF question is meaningful because visibility is measured in rendered image.

Future system can combine both representations:

1. Policy proposes semantic Gaussian TF from spoken or typed instruction.
2. Gaussian proposal converts to standard VTK piecewise-linear points.
3. Expert edits arbitrary points when more control is needed.
4. Visibility measurement evaluates final rendered result.
5. Preference or RL training learns from accepted refinements.

This hybrid keeps RL action space compact while preserving expert-level curve editing. It also gives future human-feedback training a correction signal: accepted point edits become demonstrations of how semantic policy should refine its proposal.

10. Source Files

File	Role
<code>transfer.py</code>	TF parameterization and lookup-table conversion
<code>visibility.py</code>	scan-aware class visibility and brightness measurement

File	Role
<code>goals.py</code>	instruction goals, distances, and attainment
<code>rl/ oneshot_env.py</code>	one-shot observation, action, and reward interface
<code>rl/baselines.py</code>	action encoding and search baselines
<code>server.py</code>	interactive command execution and comparison arms